

NLP

Lecture 1 - The NLP classics

Lecture:

- Overview of NLP in business analytics
- Logistics
- Review of data processing in python;
- Classic text mining techniques

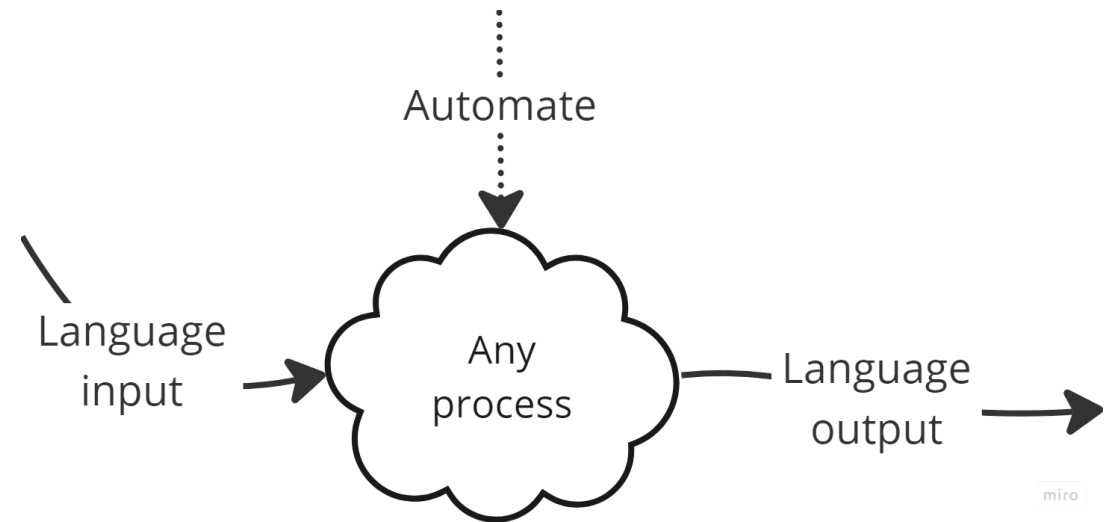
Lab:

- Setup ChatGPT
- Setup Google Colab
- Data mining on Twitter

NLP: Natural language processing

- make computers automate reading, writing and speaking

- Academia: innovation for the best algorithm for automation
- Industry: Application to generate business value
 - Data sentients: algorithm tailoring for business solution
 - Data engineer: product optimization, reliability and cost effectiveness
- Government: regulations for best societal interests



How a businessman thinks about NLP

$$\begin{aligned} \textit{Profit} &= \textit{Revenue} - \textit{Cost} \\ &= \textit{Sale Price} \times \textit{Volume} - \textit{Cost} \end{aligned}$$

- Examples

Syllabus - see Blackboard

Review of data analytics in python

- https://www.w3schools.com/python/pandas/pandas_dataframes.asp

Break and team-up

1. Find your home team. Stay with the same team throughout the course.
2. Name your team
3. Introduce yourself to your teammates
4. Each team pitches an NLP project to rest of class

Team assignment: ([link](#))

Classic text mining techniques

- Regular expressions
<https://regexr.com/>
- Regex using python
https://www.w3schools.com/python/python_regex.asp

Edit distance - aka Levenshtein distance

The ***minimum number of single-character edits*** (insertions, deletions or substitutions) required to change one word into the other

Application: Spelling error measure, OCR (optical character recognition)

Reference:

<https://phiresky.github.io/levenshtein-demo/>

https://en.wikipedia.org/wiki/Edit_distance

Example:

1. "c" vs "a"
2. "ac" vs "ca"
3. "amc" vs "cam"
4. "ymca" vs "macy"
5. "y m c a" vs "m a c y"

Definition:

$$\text{lev}(a, b) = \begin{cases} |a| & \text{if } |b| = 0, \\ |b| & \text{if } |a| = 0, \\ \text{lev}(\text{tail}(a), \text{tail}(b)) & \text{if } \text{head}(a) = \text{head}(b), \\ 1 + \min \begin{cases} \text{lev}(\text{tail}(a), b) \\ \text{lev}(a, \text{tail}(b)) \\ \text{lev}(\text{tail}(a), \text{tail}(b)) \end{cases} & \text{otherwise} \end{cases}$$

Word Error Rate

Application: ASR (automated speech recognition)

Reference:

<https://www.amberscript.com/en/resources/wer-tool/>

Example:

1. “ymca” vs “macy”
2. “y m c a” vs “m a c y”

Definition:

$$WER = \frac{S + D + I}{N} = \frac{S + D + I}{S + D + C}$$

Break

Lab 1: Set up ChatGPT

- <https://openai.com/blog/chatgpt>

Lab 2: Set up Google Colab

- <https://colab.research.google.com/>

Lab 3: Mining Twitter data

- https://colab.research.google.com/drive/18U8EFXr0TtqrfAXS2gxhh8eSj77_oAMm
- Save a copy of the notebook above
- Make a copy of the following data in your Google Drive. Note the path <https://drive.google.com/file/d/1vZs8mw6cIIEWDeJbAitfAjql6oH2J6-R/view?usp=sharing>